



5.2.1 Gaussian elimination ¶

5.2.1 Gaussian elimination

fit width



Homework 5.2.1.1. Solve

$$\begin{pmatrix} 2 & -1 & 1 \\ -4 & 0 & 1 \\ 4 & 0 & -2 \end{pmatrix} \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix} = \begin{pmatrix} -6 \\ 2 \\ 0 \end{pmatrix}.$$

► [Answer](#) ► [Solution](#)

The exercise in [Homework 5.2.1.1](#) motivates the following algorithm, which reduces the linear system $Ax = b$ stored in $n \times n$ matrix A and right-hand side vector b of size n to an upper triangular system.

```

for  $j := 0, \dots, n-1$ 
  for  $i := j+1, \dots, n-1$ 
     $\lambda_{i,j} := \alpha_{i,j} / \alpha_{j,j}$ 
     $\alpha_{i,j} := 0$ 
    for  $k = j+1, \dots, n-1$ 
       $\alpha_{i,k} := \alpha_{i,k} - \lambda_{i,j} \alpha_{j,k}$ 
    endfor
     $\beta_i := \beta_i - \lambda_{i,j} \beta_j$ 
  endfor
endfor

```

} subtract $\lambda_{i,j}$ times row j from row k

This algorithm completes as long as no divide by zero is encountered.

Let us manipulate this a bit. First, we notice that we can first reduce the matrix to an upper triangular matrix, and then update the right-hand side using the multipliers that were computed along the way (if these are stored):

```

reduce  $A$  to upper triangular form
for  $j := 0, \dots, n-1$ 
  for  $i := j+1, \dots, n-1$ 
     $\lambda_{i,j} := \alpha_{i,j} / \alpha_{j,j}$ 
     $\alpha_{i,j} := 0$ 
    for  $k = j+1, \dots, n-1$ 
       $\alpha_{i,k} := \alpha_{i,k} - \lambda_{i,j} \alpha_{j,k}$ 
    endfor
  endfor
endfor

update  $b$  using multipliers (forward substitution)
for  $j := 0, \dots, n-1$ 
  for  $i := j+1, \dots, n-1$ 
     $\beta_i := \beta_i - \lambda_{i,j} \beta_j$ 
  endfor
endfor

```

Ignoring the updating of the right-hand side (a process known as forward substitution), for each iteration we can first compute the multipliers and then update the matrix:

```

for  $j := 0, \dots, n-1$ 
  for  $i := j+1, \dots, n-1$ 
     $\lambda_{i,j} := \alpha_{i,j} / \alpha_{j,j}$ 
     $\alpha_{i,j} := 0$ 
  endfor
  for  $i := j+1, \dots, n-1$ 
    for  $k = j+1, \dots, n-1$ 
       $\alpha_{i,k} := \alpha_{i,k} - \lambda_{i,j} \alpha_{j,k}$ 
    endfor
  endfor
endfor

```

} compute multipliers

} subtract $\lambda_{i,j}$ times row j from row k

Since we know that $\alpha_{i,j}$ is set to zero, we can use its location to store the multiplier:

```

for j := 0, ..., n - 1
  for i := j + 1, ..., n - 1
     $\alpha_{i,j} := \lambda_{i,j} = \alpha_{i,j} / \alpha_{j,j}$ 
  endfor
  for k := j + 1, ..., n - 1
     $\alpha_{i,k} := \alpha_{i,k} - \alpha_{i,j} \alpha_{j,k}$ 
  endfor
endfor
endfor

```

Finally, we can cast the computation in terms of operations with vectors and submatrices:

```

for j := 0, ..., n - 1
   $\begin{pmatrix} \alpha_{j+1,j} \\ \vdots \\ \alpha_{n-1,j} \end{pmatrix} := \begin{pmatrix} \alpha_{j+1,j} \\ \vdots \\ \alpha_{n-1,j} \end{pmatrix} / \alpha_{j,j}$ 
   $\begin{pmatrix} \alpha_{j+1,j+1} & \cdots & \alpha_{j+1,n-1} \\ \vdots & & \vdots \\ \alpha_{n-1,j+1} & \cdots & \alpha_{n-1,n-1} \end{pmatrix} :=$ 
   $\begin{pmatrix} \alpha_{j+1,j+1} & \cdots & \alpha_{j+1,n-1} \\ \vdots & & \vdots \\ \alpha_{n-1,j+1} & \cdots & \alpha_{n-1,n-1} \end{pmatrix} - \begin{pmatrix} \alpha_{j+1,j} \\ \vdots \\ \alpha_{n-1,j} \end{pmatrix} \begin{pmatrix} \alpha_{j,j+1} & \cdots & \alpha_{j,n-1} \end{pmatrix}$ 
endfor

```

In [Figure 5.2.1.1](#) this algorithm is presented with our FLAME notation.

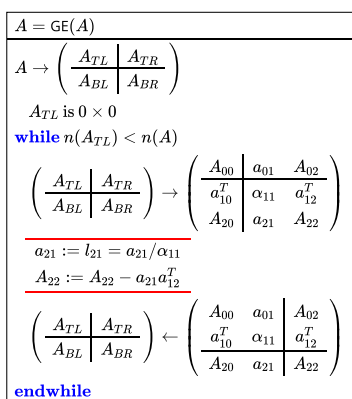


Figure 5.2.1.1. Gaussian elimination algorithm that reduced a matrix A to upper triangular form, storing the multipliers below the diagonal.

Homework 5.2.1.2. Apply the algorithm [Figure 5.2.1.1](#) to the matrix

$$\begin{pmatrix} 2 & -1 & 1 \\ -4 & 0 & 1 \\ 4 & 0 & -2 \end{pmatrix}$$

and report the resulting matrix. Compare the contents of that matrix to the upper triangular matrix computed in the solution of [Homework 5.2.1.1](#).

[Answer](#) [Solution](#)

Homework 5.2.1.3. Applying [Figure 5.2.1.1](#) to the matrix

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -4 & 0 & 1 \\ 4 & 0 & -2 \end{pmatrix}$$

yielded

$$\begin{pmatrix} 2 & -1 & 1 \\ -2 & -2 & 3 \\ 2 & -1 & -1 \end{pmatrix}.$$

This can be thought of as an array that stores the unit lower triangular matrix L below the diagonal (with implicit ones on its diagonal) and upper triangular matrix U on and above its diagonal:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 2 & -1 & 1 \\ 0 & -2 & 3 \\ 0 & 0 & -1 \end{pmatrix}$$

Compute $B = LU$ and compare it to A .

[Answer](#) [Solution](#)

$$B = LU = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 1 \\ 0 & -2 & 3 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 1 \\ -4 & 0 & 1 \\ 4 & 0 & -2 \end{pmatrix} = A.$$

Magic! $B = A$!